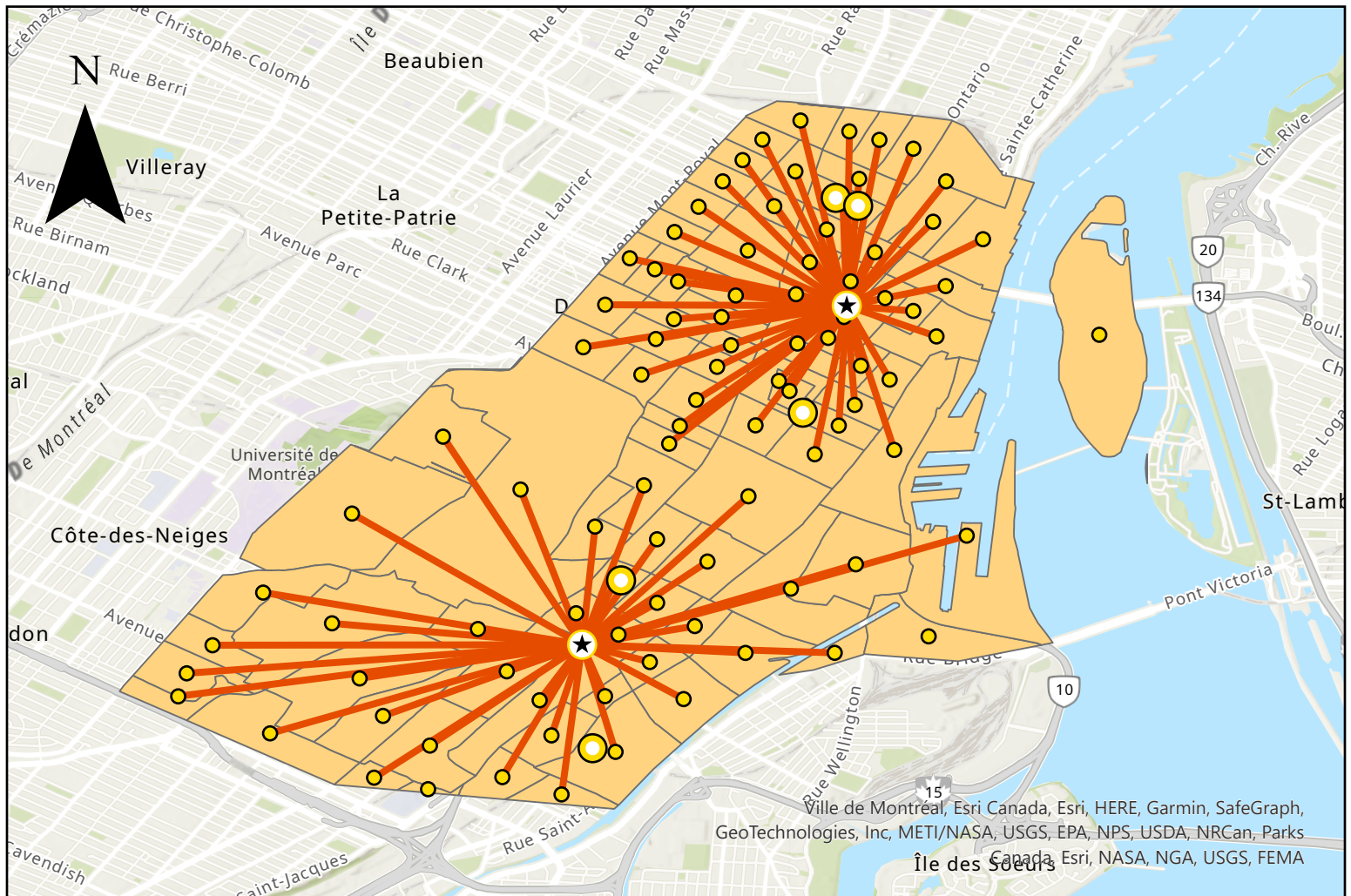


2 Selected Montreal CLSCs to Maintain Based on Population Distributions



1 0.5 0 1 2 3 4
Kilometers

Legend

Facilities

- ★ Chosen
- Required
- Competitor
- Candidate
- Demand Points
- Line Between Demand Points and Chosen CLSCs
- Census Tracts

Data Source: GEOG 414

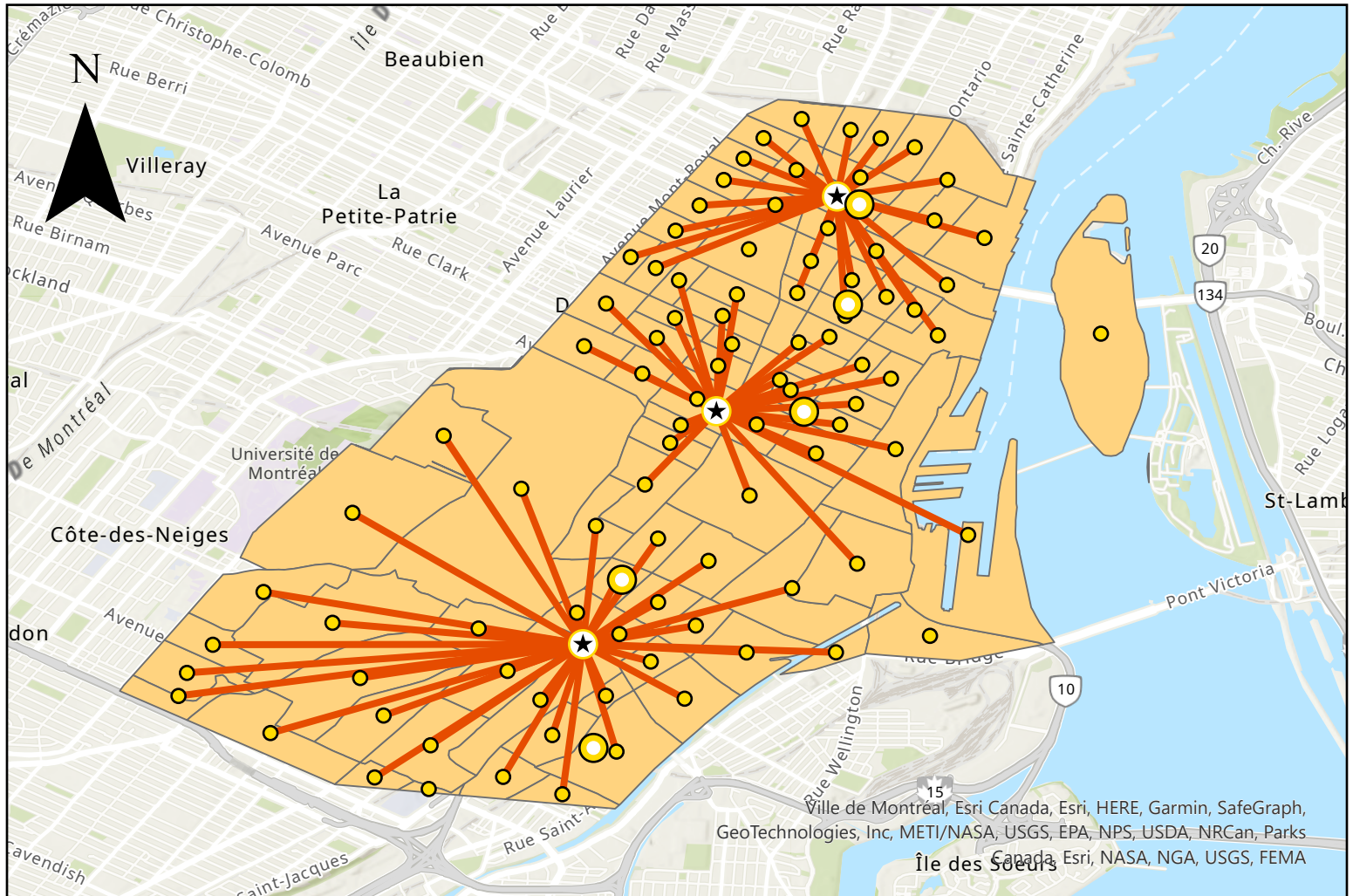
Projected Coordinate System: NAD 1983 UTM Zone 18N

Author: Grady Zappone

Date: 18 September 2023

A map displaying the 2 primary health care facilities in downtown Montreal to be maintained to minimize impacts on accessibility based on population distributions.

3 Selected Montreal CLSCs to Maintain Based on Population Distributions



Legend

Facilities

- ★ Chosen
- Required
- Competitor
- Candidate
- Demand Points
- Line Between Demand Points and Chosen CLSCs
- Census Tracts

1 0.5 0 1 2 3 4
Kilometers

Data Source: GEOG 414

Projected Coordinate System: NAD 1983 UTM Zone 18N

Author: Grady Zappone

Date: 18 September 2023

A map displaying the three primary health care facilities in downtown Montreal to be maintained to maximize accessibility based on population distributions. One of the chosen candidates is to be built new, while the other two are existing CLSCs.

Report on the Evaluation of CLSC Facilities in Downtown Montreal

This report presents the findings of an analysis aimed at assessing the distribution of CLSC (Centre Local de Services Communautaire) facilities in downtown Montreal. The first objective of this assessment was to provide a recommendation for two CLSCs (out of seven total) that should remain open due to cost-cutting measures. To evaluate this, the centroids of each census tract polygon represented a demand point with a weight corresponding to its population. Using a location allocation analysis, each census tract is assigned to the closest CLSC. When the number of CLSCs to select was restricted to two, Location 1 and Location 6 were selected as the optimal locations to remain open. This is because they were found to be strategically positioned to serve the highest population densities within the downtown Montreal area, ensuring continued access to essential healthcare services for the greatest number of residents in the closest location. Location 1 services 38 census tracts primarily on the southwestern half of downtown Montreal, with a combined population of 102,152 people. On the other hand, Location 6 serves a larger number of census tracts, primarily located in the northeast side of downtown Montreal, but these tracts contain a smaller population of 99,580 people. Ultimately, the selection of Location 1 and Location 6 was a data-driven choice, balancing the geographical distribution of service coverage and population served to optimize the allocation of healthcare resources while adhering to budget constraints.

The second objective of this analysis was to choose an additional site to construct a CLSC that may better serve the needs of the population. I selected the new candidate CLSC to be located centrally in downtown Montreal. With this new CLSC candidate and the capacity to choose 3 optimal CLSCs, the location allocation analysis selected Location 1, Location 3 the new location. In this scenario, Location 1 now serves 34 census tracts, with a total population of 93,096 people. Location 3 serves 28 census tracts in northeast of downtown Montreal, containing 50,251 people. The new location serves 28 census tracts that hold 58,385 people. The addition of the new CLSC decreases the burden on Location 1 by 9,056 people. It also splits the load that was on Location 6 in the last scenario onto Location 3 and the new location. This strategic reconfiguration could yield substantial cost savings without significantly compromising healthcare accessibility for the local population.